Feed

Feed is the relative movement of the work or tool in a direction perpendicular to the axis of reciprocation of the ram per double stroke. It is expressed in mm per stroke.

3.17 Coolant

Due to the friction between the tool and the work surface during machining, heat is generated. The tool loses its cutting capacity and the machined surface is hardened. Coolant is used on the surfaces to avoid damage to the cutting edge of the tool as well as to the machined surface.

Soluble oil is mixed with water to be used as a suitable coolant. One part of the oil is mixed with fifteen parts of water to be used as coolant. Usage of water as coolant may result in rust formation on the metal parts. Lubricants cannot be used as coolants.

3.18 Safety precautions

The following safety precautions should be observed while working on a shaping machine.

Safety precautions regarding operators

1. No alteration or adjustment should be done on the machine parts while the machine is functioning.
2. Clamps holding the work should not be adjusted while the machine is in operation.
3. The machine is to be stopped before cleaning the metal chips.
4. The sharp edges of the work should be handled with care.
5. The measuring of the work should be done only after the machine is switched off.
6. The operator should not seek the assistance of others for starting and stopping the machine.
7. Machining of precision parts and internal surfaces of the workpiece are to be carried out with great care and attention.
8. The operator should stay away from direction of the ram movement.

Safety hints regarding the shaping machine

1. The workpiece is to be positioned in such a way that ram will not hit the workpiece while performing the forward stroke.
2. Strokelength of the ram and the position of stroke are to be set correctly before performing the operation.
3. Proper holding of the work should be ensured. Work holding devices like clamps and vise jaws should not come in the way of the reciprocating tool.
4. We have to ensure that the tool or the tool post or the ram will not hit the job or the job holding clamps or the vise jaws.
5. The machine should be stopped before making any adjustment to the strokelength, position of stroke, apron and tool position.

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